Jinnuo Liu

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Research: Representation learning; LLM evaluation and post-training

EDUCATION

New York University, Courant Institute 2026 – Present
Ph.D. in Computer Science · Advisor: Prof. Hongyi Wen

New York University Shanghai 2026
B.S. in Computer Science; Minor in Interactive Media Arts GPA: 3.888
Magna cum laude · NYU Shanghai Excellence Award
Major Honors in Computer Science, Data Science, and Engineering (Computer Science)

PUBLICATIONS

* Equal contribution.

[1] **Jinnuo Liu***, Junhao Zhu*, Weifeng Jiang, Haoming Liu, Hongyi Wen. **Coherence-Aware Distributional Evaluation of Open-Ended Text Generation**. 2026. arXiv:2609.34240.

[2] **Jinnuo Liu**, Yue Peng, Jinhan Niu, Hongyi Wen. **NovelAPIBench: Diagnosing How A Code LLM Learns to Use Novel APIs**. 2026. arXiv:2606.03657.

[3] Haoming Liu, **Jinnuo Liu**, Yanhao Li, Liuyang Bai, Yunkai Ji, Yuanhe Guo, Shenji Wan, Hongyi Wen. **From Navigation to Refinement: Revealing the Two-Stage Nature of Flow-based Diffusion Models through Oracle Velocity**. *CVPR 2026, Findings track*. arXiv:2512.02826.

[4] Yuantao Deng*, **Jinnuo Liu***, Kaizhen Tan, Yuchen Liu. **Early Learning Shapes Later Directions of Representation Change in Continual Learning**. 2026. arXiv:2609.36081.

[5] **Jinnuo Liu**, Chuke Liu, Hua Shen. **ValueFlow: Measuring the Propagation of Value Perturbations in Multi-Agent LLM Systems**. 2026. arXiv:2602.08567.

INDUSTRY EXPERIENCE

AI Engineer Intern | **China Literature, Tencent PCG** May 2025 – Aug 2025

- Developed and deployed RAG pipelines, data processing workflow, and backend services for customer-support applications serving Qidian and QQ Reading apps.
- Contributed to evaluation pipelines and optimized service latency and stability under concurrent workloads.

SELECTED RESEARCH EXPERIENCE

CHORD: Coherence-Aware Generation Evaluation Jun 2026 – Sep 2026
Lead researcher; co-first author · Publication [1]

- Identified global coherence failures missed by perplexity, unigram entropy, and MAUVE; led problem formulation, experiments, and writing for a coherence-aware distributional metric.
- Detected **9/9 coherence failure types** at highest severity against benign rewrites; achieved **Spearman** $\rho = 0.98$ with human rankings of sensibility and human-likeness across eight GPT-2 generation settings.
- Contributed to distilling the **27B evaluator into 0.8B**, reducing feature-extraction time from **80.1s to 9.0s** and peak GPU memory from **54.0GB to 4.2GB** on 700 documents with a single H200.

NovelAPIBench: Diagnosing Novel API Acquisition Mar 2026 – Sep 2026
Lead researcher; first author · Publication [2]

- Led framework design, implementation, experiments, and writing; built a diagnostic benchmark with **1,670 tasks across 856 APIs from 20 libraries** in its primary instance and evaluated **six code LLMs**.
- Showed that SFT/RAFT improve external-knowledge use: Pass@1 rose from **71.3% to 82.8%/85.3%** with API knowledge on **348 held-out tasks**, but remained at or below **2.3%** without it.

OOD-Aware On-Policy Self-Distillation Ongoing
Research collaborator · Manuscript in preparation

- Implementing and evaluating self-distillation baselines, diagnosing failure modes, and co-developing a simpler method to improve generalization to out-of-domain tasks.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, Bash.
ML frameworks: PyTorch, Hugging Face Transformers, Accelerate, Diffusers, LangChain, scikit-learn.
Data & visualization: NumPy, pandas, Matplotlib.
Systems & tools: High-performance computing (HPC), Git, Docker, LaTeX.
AI-assisted development: Claude Code, Codex.